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CS3391 - Object Oriented Programming

Topic: super keyword

1. SUMMARY

The super keyword in Object Oriented Programming (OOP) is used to inherit properties and methods from a parent class. It is a fundamental concept in OOP that enables code reuse and facilitates the creation of complex class hierarchies. The super keyword allows a subclass to call the overridden method from the parent class, providing a way to access the parent class's implementation. In the context of Java, the super keyword is used to invoke the constructor of the parent class, ensuring that the parent class's initialization is performed before the subclass's initialization. The super keyword also enables the use of polymorphism, where objects of different classes can be treated as objects of a common superclass.

The super keyword is a key feature of OOP, enabling the creation of complex class hierarchies and facilitating code reuse. It is used in various contexts, including inheritance, polymorphism, and object initialization. The super keyword has several advantages, including code reuse, reduced complexity, and improved maintainability.

2. SHORT NOTES

Introduction to Super Keyword

The super keyword is a fundamental concept in Object Oriented Programming (OOP) that enables the inheritance of properties and methods from a parent class. It is used to call the overridden method from the parent class, providing a way to access the parent class's implementation.

Types and Classifications

There are two main types of super keyword usage:

1. **Invoking the constructor of the parent class**: The super keyword is used to invoke the constructor of the parent class, ensuring that the parent class's initialization is performed before the subclass's initialization.
2. **Calling the overridden method from the parent class**: The super keyword is used to call the overridden method from the parent class, providing a way to access the parent class's implementation.

Working Principle

The working principle of the super keyword is as follows:

1. The subclass inherits the properties and methods of the parent class.
2. The subclass can override the methods of the parent class by declaring a new method with the same name and signature.
3. The subclass can use the super keyword to invoke the constructor of the parent class.
4. The subclass can use the super keyword to call the overridden method from the parent class.

Important Formulas and Equations

There are no specific formulas or equations associated with the super keyword.

Diagrams Description

A simple class hierarchy diagram can be used to illustrate the concept of inheritance and the use of the super keyword.

Advantages and Disadvantages

Advantages:

3. PRACTICAL / CODING EXAMPLES

* Reduced complexity
* Improved maintainability

Disadvantages:

* None

Real World Applications

The super keyword has numerous applications in real-world programming, including:

* Creating complex class hierarchies
* Facilitating code reuse
* Improving maintainability

Comparison with Related Concepts

The super keyword is closely related to the concept of inheritance and polymorphism.

Time and Space Complexity

The time and space complexity of the super keyword is O(1) in terms of time complexity and O(1) in terms of space complexity.

Important Terminology

* **Super keyword**: A keyword used to invoke the constructor of the parent class or call the overridden method from the parent class.
* **Parent class**: A class that is inherited by another class.
* **Subclass**: A class that inherits the properties and methods of a parent class.

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Based on the subject name above, this is a hands-on implementation subject. Therefore, we will provide code examples in C and Python.

C Implementation (CORE LOGIC ONLY, NO main function, NO headers)

```c
void Parent::parentMethod() {
    printf("Parent method called\n");
}

void Child::childMethod() {
    super.parentMethod(); // Call the overridden method from the parent class
}

void Child::childConstructor() {
    super.parentConstructor(); // Invoke the constructor of the parent class
}
```

Python Implementation (CORE LOGIC ONLY)

```python
class Parent:
    def parentMethod(self):
        print("Parent method called")

class Child(Parent):
    def childMethod(self):
        super().parentMethod()  # Call the overridden method from the parent class
```

```
    def childConstructor(self):
        super().__init__()  # Invoke the constructor of the parent class
```

Algorithm (Brief step by step)

1. Declare the parent class with its properties and methods.
2. Declare the subclass with its properties and methods.
3. Use the super keyword to invoke the constructor of the parent class.
4. Use the super keyword to call the overridden method from the parent class.

Time and Space Complexity

Time complexity: O(1)

Space complexity: O(1)

4. IMPORTANT QUESTIONS

Part-A (2 Marks) ? 5 questions, each with a crisp 2-3 line answer:

Q1. Define the super keyword in Java and its purpose.

Answer: The super keyword in Java is used to refer to the parent class from a child class. It is used to access the members of the parent class.

Q2. Compare the super keyword with the this keyword in Java.

Answer: The super keyword is used to access the parent class members, while the this keyword is used to access the current class members.

Q3. What is the purpose of using the super keyword in method overriding?

Answer: The super keyword is used to call the parent class method from a child class method, allowing for method overriding.

Q4. Write a short note on the use of the super keyword in constructor chaining.

Answer: The super keyword is used in constructor chaining to call the parent class constructor from a child class constructor.

Q5. Give an example of using the super keyword in a real-world application.

Answer: The super keyword can be used in a real-world application such as a banking system, where a child class "SavingsAccount" can use the super keyword to access the parent class "Account" members.

Part-B (13 Marks) ? 3 questions. Each answer MUST be at least 20-25 lines long.

Q1. Explain super keyword in detail with neat diagram and suitable examples.

Answer:

The super keyword in Java is a reference variable that is used to refer to the parent class from a child class. It is used to access the members of the parent class, such as methods and variables. The super keyword is particularly useful in method overriding, where a child class provides a different implementation of a method that is already defined in its parent class.

There are several types of super keyword, including:

* super(): This is used to call the parent class constructor from a child class constructor.
* super.variable: This is used to access a variable of the parent class from a child class.
* super.method(): This is used to call a method of the parent class from a child class.

Here is a diagram that illustrates the use of the super keyword:

```
  +---------------+
  |  Parent Class  |
  +---------------+
          |
          |
```

```
        v
  +---------------+
  |  Child Class   |
  |  (uses super)  |
  +---------------+
```

The diagram shows that the child class uses the super keyword to access the members of the parent class.

The working of the super keyword can be explained in the following steps:

1. The child class inherits the members of the parent class using the extends keyword.
2. The child class uses the super keyword to access the members of the parent class.
3. The super keyword is used to call the parent class constructor from a child class constructor.
4. The super keyword is used to access a variable of the parent class from a child class.
5. The super keyword is used to call a method of the parent class from a child class.

The super keyword has several advantages, including:

* It allows for code reuse by accessing the members of the parent class.
* It allows for method overriding by providing a different implementation of a method that is already defined in the parent class.
* It allows for constructor chaining by calling the parent class constructor from a child class constructor.

However, the super keyword also has some disadvantages, including:

* It can lead to tight coupling between the parent and child classes.
* It can make the code harder to understand and maintain.

The super keyword has several real-world applications, including:

* Banking system: A child class "SavingsAccount" can use the super keyword to access the parent class "Account" members.
* Employee management system: A child class "Manager" can use the super keyword to access the parent class "Employee" members.
* University management system: A child class "Professor" can use the super keyword to access the parent

class "Teacher" members.

In conclusion, the super keyword is an important concept in Java that allows for code reuse and method overriding. It has several advantages and disadvantages, and it has several real-world applications.

Q2. Describe the types of super keyword with working principle and examples.

Answer:

There are several types of super keyword in Java, including:

* **super()**: This is used to call the parent class constructor from a child class constructor.
* **super.variable**: This is used to access a variable of the parent class from a child class.
* **super.method()**: This is used to call a method of the parent class from a child class.

Here is an example of using the super() keyword:

```java
class Parent {
    Parent() {
        System.out.println("Parent class constructor");
    }
}

class Child extends Parent {
    Child() {
        super(); // calls the parent class constructor
        System.out.println("Child class constructor");
    }
}
```

Here is an example of using the super.variable keyword:

```java
class Parent {
    int x = 10;
```

```
}

class Child extends Parent {
    void display() {
        System.out.println("Value of x in parent class: " + super.x);
    }
}
```

Here is an example of using the super.method() keyword:

```java
class Parent {
    void display() {
        System.out.println("Parent class method");
    }
}

class Child extends Parent {
    void display() {
        super.display(); // calls the parent class method
        System.out.println("Child class method");
    }
}
```

The working principle of the super keyword can be explained in the following steps:

1. The child class inherits the members of the parent class using the extends keyword.
2. The child class uses the super keyword to access the members of the parent class.
3. The super keyword is used to call the parent class constructor from a child class constructor.
4. The super keyword is used to access a variable of the parent class from a child class.
5. The super keyword is used to call a method of the parent class from a child class.

Here is a comparison table of the different types of super keyword:

Type	Description	Example
super()	Calls the parent class constructor	super()
super.variable	Accesses a variable of the parent class	super.x
super.method()	Calls a method of the parent class	super.display()

The super keyword has several real-world applications, including:

* Banking system: A child class "SavingsAccount" can use the super keyword to access the parent class "Account" members.

* Employee management system: A child class "Manager" can use the super keyword to access the parent class "Employee" members.

* University management system: A child class "Professor" can use the super keyword to access the parent class "Teacher" members.

In conclusion, the super keyword is an important concept in Java that allows for code reuse and method overriding. It has several types, including super(), super.variable, and super.method(), and it has several real-world applications.

Q3. Compare super keyword with related concepts. Discuss advantages and applications.

Answer:

The super keyword is related to several other concepts in Java, including:

* **this keyword**: The this keyword is used to refer to the current object, while the super keyword is used to refer to the parent class.

* **inheritance**: Inheritance is the mechanism by which a child class inherits the members of a parent class.

* **method overriding**: Method overriding is the mechanism by which a child class provides a different implementation of a method that is already defined in its parent class.

Here is a comparison table of the super keyword with related concepts:

Concept	Description	Example

| super keyword | Refers to the parent class | super() |

| this keyword | Refers to the current object | this.x |

| inheritance | Mechanism by which a child class inherits the members of a parent class | class Child extends Parent |

| method overriding | Mechanism by which a child class provides a different implementation of a method that is already defined in its parent class | void display() { super.display(); } |

The super keyword has several advantages over related concepts, including:

* **code reuse**: The super keyword allows for code reuse by accessing the members of the parent class.
* **method overriding**: The super keyword allows for method overriding by providing a different implementation of a method that is already defined in the parent class.
* **constructor chaining**: The super keyword allows for constructor chaining by calling the parent class constructor from a child class constructor.

The super keyword has several applications, including:

* **banking system**: A child class "SavingsAccount" can use the super keyword to access the parent class "Account" members.
* **employee management system**: A child class "Manager" can use the super keyword to access the parent class "Employee" members.
* **university management system**: A child class "Professor" can use the super keyword to access the parent class "Teacher" members.

In conclusion, the super keyword is an important concept in Java that allows for code reuse and method overriding. It has several advantages over related concepts, including code reuse, method overriding, and constructor chaining, and it has several real-world applications.

Part-C (15 Marks) ? 2 questions. Each answer MUST be at least 30 lines long.

Q1. Elaborate on super keyword covering theory, implementation, real-world applications, and future scope.

Answer:

The super keyword is a fundamental concept in Java that allows for code reuse and method overriding. It is used to refer to the parent class from a child class, and it is particularly useful in method overriding, where a child class provides a different implementation of a method that is already defined in its parent class.

Section 1: Deep Theoretical Background

The super keyword is based on the concept of inheritance, which is the mechanism by which a child class inherits the members of a parent class. Inheritance is a fundamental concept in object-oriented programming, and it allows for code reuse and modularity. The super keyword is used to access the members of the parent class, such as methods and variables, from a child class.

The super keyword is also related to the concept of method overriding, which is the mechanism by which a child class provides a different implementation of a method that is already defined in its parent class. Method overriding is used to provide a specific implementation of a method for a particular class, while still allowing for code reuse and modularity.

Section 2: How it is Implemented/Used in Practice

The super keyword is implemented in Java using the extends keyword, which is used to inherit the members of a parent class. The super keyword is used to access the members of the parent class, such as methods and variables, from a child class. The super keyword can be used in several ways, including:

* **super()**: This is used to call the parent class constructor from a child class constructor.
* **super.variable**: This is used to access a variable of the parent class from a child class.
* **super.method()**: This is used to call a method of the parent class from a child class.

Here is an example of using the super keyword:

```java
class Parent {
    void display() {
        System.out.println("Parent class method");
    }
}

class Child extends Parent {
```

```
    void display() {
        super.display(); // calls the parent class method
        System.out.println("Child class method");
    }
}
```

Section 3: Real-World Applications

The super keyword has several real-world applications, including:

* **banking system**: A child class "SavingsAccount" can use the super keyword to access the parent class "Account" members.
* **employee management system**: A child class "Manager" can use the super keyword to access the parent class "Employee" members.
* **university management system**: A child class "Professor" can use the super keyword to access the parent class "Teacher" members.

Here is a detailed case study of a banking system:

* The banking system has a parent class "Account" that has members such as account number, account holder name, and balance.
* The banking system has a child class "SavingsAccount" that inherits the members of the parent class "Account" and adds additional members such as interest rate and minimum balance.
* The child class "SavingsAccount" uses the super keyword to access the members of the parent class "Account", such as account number and account holder name.

Section 4: Limitations and Challenges

The super keyword has several limitations and challenges, including:

* **tight coupling**: The super keyword can lead to tight coupling between the parent and child classes, which can make the code harder to maintain and modify.
* **complexity**: The super keyword can make the code more complex, especially when there are multiple levels of inheritance.

Section 5: Future Scope

The super keyword has several future scope, including:

* **improved code reuse**: The super keyword can be used to improve code reuse by accessing the members of the parent class.

* **improved modularity**: The super keyword can be used to improve modularity by providing a specific implementation of a method for a particular class.

In conclusion, the super keyword is an important concept in Java that allows for code reuse and method overriding. It has several real-world applications, including banking system, employee management system, and university management system. However, it also has several limitations and challenges, including tight coupling and complexity.

Q2. Critically analyze super keyword. Compare approaches, evaluate trade-offs, and justify with examples.

Answer:

The super keyword is a fundamental concept in Java that allows for code reuse and method overriding. It is used to refer to the parent class from a child class, and it is particularly useful in method overriding, where a child class provides a different implementation of a method that is already defined in its parent class.

Section 1: Overview of the Concept and its Importance

The super keyword is based on the concept of inheritance, which is the mechanism by which a child class inherits the members of a parent class. Inheritance is a fundamental concept in object-oriented programming, and it allows for code reuse and modularity. The super keyword is used to access the members of the parent

5. MCQs

Q1. What is the primary purpose of the super keyword in OOP?

a) To invoke the constructor of the parent class

b) To call the overridden method from the parent class

c) To create a new class hierarchy

d) To improve code maintainability

Answer: b) To call the overridden method from the parent class

Explanation: The super keyword is used to call the overridden method from the parent class, providing a way to access the parent class's implementation.

Q2. What is the advantage of using the super keyword in OOP?

a) Code reuse

b) Reduced complexity

c) Improved maintainability

d) All of the above

Answer: d) All of the above

Explanation: The super keyword enables code reuse, reduces complexity, and improves maintainability.

Q3. What is the time complexity of the super keyword in OOP?

a) O(1)

b) O(n)

c) O(log n)

d) O(n^2)

Answer: a) O(1)

Explanation: The time complexity of the super keyword is O(1), meaning it takes constant time to execute.

Q4. What is the space complexity of the super keyword in OOP?

a) O(1)

b) O(n)

c) O(log n)

d) O(n^2)

Answer: a) O(1)

Explanation: The space complexity of the super keyword is O(1), meaning it takes constant space to execute.

Q5. What is the purpose of the super keyword in the context of Java?

a) To invoke the constructor of the parent class

b) To call the overridden method from the parent class

c) To create a new class hierarchy

d) To improve code maintainability

Answer: a) To invoke the constructor of the parent class

Explanation: In Java, the super keyword is used to invoke the constructor of the parent class, ensuring that the parent class's initialization is performed before the subclass's initialization.

7. YOUTUBE LINKS

- [Super Keyword in OOP Tutorial](https://www.youtube.com/results?search_query=super+keyword+oop+tutorial)
- [Inheritance and Polymorphism in OOP](https://www.youtube.com/results?search_query=inheritance+polymorphism+oop+tutorial)
- [Object Oriented Programming Tutorial](https://www.youtube.com/results?search_query=oop+tutorial)
- [Java OOP Tutorial](https://www.youtube.com/results?search_query=java+oop+tutorial)
- [Super Keyword in Java Tutorial](https://www.youtube.com/results?search_query=super+keyword+java+tutorial)